Exercises 3.6 3, 5, 7, 9, 13, 15, 19, 33, 39, 41, 47, 56.

3. $f(x) = \ln(x^3 + 3x + 3)$

$$f'(x) = \frac{1}{x^3+3x+3} \cdot (3x^2+3)$$

5. $f(x) = \sin(\ln x)$

$$f'(x) = \cos(\ln x) \cdot \frac{1}{x}$$

7. $f(x) = \ln \frac{1}{x}$

$$f'(x) = x \cdot (-x^{-2})$$

$$= -x \cdot \frac{1}{x^2}$$

$$= -\frac{1}{x}$$

9. $g(x) = \ln(x e^{-2x})$

$$g'(x) = \frac{1}{x \cdot e^{-2x}} \cdot \left[e^{-2x} + x e^{-2x} \cdot (-2)\right]$$

$$= \frac{1-2x}{x} \qquad = \frac{1}{x} - 2$$

Exercises 3.6 3, 5, 7, 9, 13, 15, 19, 33, 39, 41, 47, 56.

13. $y = \log_8(x^2+3x)$

$$y' = \frac{1}{(x^2+3x)\ln 8} \cdot (2x+3)$$

$$= \frac{2x+3}{(x^2+3x)\ln 8}$$

15. $F(s) = \ln(\ln s)$

$$F'(s) = \frac{1}{\ln s} \cdot \frac{1}{s}$$

19. $y = \ln|3-2x^5|$

$$y' = \frac{1}{3-2x^5}(-10x^4)$$

$$= \frac{-10x^4}{3-2x^5}$$

Exercises 3.6 3, 5, 7, 9, 13, 15, 19, 33, 39, 41, 47, 56.

53. domain : $x-1>0$

$x>1$ and

$1-\ln(x-1) \neq 0$

$x \neq e+1$

∴ domain $x \in (1, e+1) \cup (e+1, \infty)$

domain(f)

$$f(x) = \frac{x}{1-\ln(x-1)}$$

$\frac{vu' - v'u}{v^2}$

$$f'(x) = \frac{[1-\ln(x-1)]\cdot 1 - x\left(\frac{1}{x-1}\right)}{[1-\ln(x-1)]^2}$$

$$= \frac{1-\ln(x-1) - \frac{x}{x-1}}{(1-\ln(x-1))^2}$$

Exercises 3.6 3, 5, 7, 9, 13, 15, 19, 33, 39, 41, 47, 56.

$y = \ln(x^2 - 3x + 1)$

$y' = \frac{1}{x^2 - 3x + 1}(2x - 3)$

$y' = \frac{2x - 3}{x^2 - 3x + 1}$

$y'(3) = \frac{2 \times 3 - 3}{3^2 - 3 \times 3 + 1} = 3$

$y - 0 = 3(x - 3)$

$y = 3x - 9$

47. $y=\sqrt{\frac{x-1}{x^4+1}}$ 可以用 ln 去做比较简单

$$y'=\frac{1}{2}\left(\frac{x-1}{x^4+1}\right)^{-\frac{1}{2}}\cdot\frac{(x^4+1)\cdot 1-(x-1)4x^3}{(x^4+1)^2}$$

$$\ln y=\frac{1}{2}\ln\left(\frac{x-1}{x^4+1}\right)$$

$$\ln y=\frac{1}{2}\ln(x-1)-\frac{1}{2}\ln(x^4+1)$$

$$\frac{1}{y}\cdot y'=\frac{1}{2(x-1)}-\frac{1}{2(x^4+1)}\Rightarrow y'=y\left(\frac{1}{2(x-1)}-\frac{1}{2(x^4+1)}\right)$$

56. $y=(\ln x)^{\cos x}$

$$\ln y=\cos x\ln(\ln x)$$

$$\frac{1}{y}\cdot y'=-\sin x\ln(\ln x)+\cos x\frac{1}{\ln x}\cdot\frac{1}{x}$$

$$\frac{1}{y}\cdot y'=\frac{\cos x}{x\ln x}-\sin x\ln(\ln x)$$